

Guangyu Meng

Ph.D. Candidate, University of Notre Dame

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RESEARCH VISION

I am building a **topology-native AI** research program founded on a fundamental observation in medical image analysis: biological structures—cells, tissues, and whole organisms—encode structural identity through topological properties (connectivity, loops, and voids), yet conventional neural networks provide no principled mechanism to capture them. My program closes this gap through three coordinated thrusts, with the long-term goal of **topology-driven computational phenomics**: trustworthy, auditable AI that characterizes all cell types and organ systems from 3D imaging data at cellular resolution.

Thrust 1: Topology-Integrated Deep Learning. Embedding persistent homology into self-supervised and unsupervised learning pipelines to produce structurally aware representations for medical image analysis (TopoCL, VTRL-Med).

Thrust 2: Scalable Topological Computation. Designing GPU-accelerated, theoretically grounded algorithms—with provable correctness and stability guarantees—for computing and comparing topological summaries at the scale demanded by phenomics (CuPH, MS-COOT, Reeb GW, NNS-EMD).

Thrust 3: Neuro-Symbolic Topological Reasoning. Building LLM-based agentic frameworks that automate topology descriptor selection across diverse imaging modalities, enabling scalable, non-expert deployment of topological analysis (TopoAgent).

I additionally contribute to efficient AI training methodology—curriculum learning for LLMs (PUDF)—enabling topology-aware models to scale to large, heterogeneous medical datasets.

EDUCATION

University of Notre Dame

Ph.D. in Computer Science (GPA: 3.92/4.0)

South Bend, IN

2021 – Present (Expected Fall 2026)

– Advisors: Prof. Danny Z. Chen (**IEEE Fellow, ACM Distinguished Scientist**) & Prof. Erin W. Chambers

University of Notre Dame

M.S. in Computer Science

South Bend, IN

August 2025

Washington University in St. Louis

Graduate Studies in Computer Science (GPA: 4.0/4.0)

St. Louis, MO

2020 – 2021

– Transferred to the Ph.D. program at the University of Notre Dame.

University of Southampton

M.S. in Microelectronics System Design (With Merit)

Southampton, UK

2012

University of Birmingham

B.Eng. in Electronic Engineering, Honours Class II (Division I)

Birmingham, UK

2010 – 2011

Huazhong University of Science and Technology

B.Eng. in Electronic Science and Engineering

Wuhan, China

2007 – 2010

RESEARCH EXPERIENCE

Graduate Research Assistant

Department of Computer Science and Engineering

University of Notre Dame

2021 – Present

Thrust 1: Topology-Integrated Deep Learning

- **TopoCL – Topological Contrastive Learning:** Designed a universal framework featuring topology-aware augmentations, a Hierarchical Transformer-based Topology Encoder, and an adaptive mixture-of-experts (MoE) fusion module for integrating visual and topological representations. Achieved +3.26% average accuracy improvement with strong statistical significance (80% of comparisons with $p < 0.001$) across five contrastive methods on five medical imaging datasets.
- **VTRL-Med – Medical Image Deep Clustering:** Developed a visual-topology framework for unsupervised medical image clustering, integrating SAM with mask-based structural priors and persistent homology features for anatomically consistent representations. Achieved state-of-the-art 81.51% accuracy across 13 benchmarks (+4.48% over baselines), nearly matching supervised methods.

Thrust 2: Scalable Topological Computation

- **CuPH – GPU-Parallel Cubical Persistent Homology:** Developed a GPU method that computes cubical persistent homology directly on pixels via level-synchronous component tracking, reducing parallel depth from $\Theta(n)$ to $O(L \log n)$ and proven to match standard cubical PH exactly on integer inputs. Achieved $18.6\text{--}99.6\times$ speedup over the prior baseline and, as a drop-in backend, accelerated topology-aware classification by $4.2\text{--}16.7\times$ and segmentation by $8.3\text{--}44\times$ while preserving accuracy.
- **MS-COOT – Morse–Smale Complex Comparison:** Developed a co-optimal transport distance for scientific visualization that models each Morse–Smale complex as a measure hypernetwork and jointly matches critical points and regions. Produces the first explicit region-to-region correspondences, enabling interpretable detection of region-level events such as splitting and merging in time-varying scalar fields that existing distances (WD, GWD, FGW) miss.
- **Reeb Graph Comparison – Computational Topology:** Proposed a provably stable framework for Reeb graph comparison using Gromov–Wasserstein distance with persistence images, establishing rigorous theoretical stability bounds under scalar field perturbations. Achieved superior classification accuracy over bottleneck distance and graph edit distance baselines.
- **NNS-EMD – Efficient Optimal Transport:** Proposed a GPU-accelerated Nearest Neighbor Search approximation for Earth Mover’s Distance with comprehensive theoretical analysis of time complexity and error bounds. Achieved $44\times\text{--}135\times$ speedup while improving accuracy (91.88% vs. 87.42% on 20news, 93.24% vs. 78.53% on Amazon reviews).

Thrust 3: Neuro-Symbolic Topological Reasoning

- **TopoAgent – Agentic Topology Learning:** Developed an LLM-based agentic framework that automates topology learning for medical image analysis using a Perception–Reasoning–Action–Reflection loop with 21 domain-specific tools, skill distillation from 15 topological descriptors across 26 datasets, and dual memory. Introduced TopoBenchmark (113,182 samples, 26 datasets, 5 object types). Achieved 68.21% average balanced accuracy, outperforming the strongest single-descriptor baseline by 9.32% and general-purpose LLMs by over 21%.

Supporting Methodology: Efficient AI Training

- **PUDF – Curriculum Learning for LLMs:** Developed a psychology-based curriculum learning framework applying Item Response Theory to artificial crowds for model-independent difficulty quantification and adaptive training scheduling. Achieved 4.13% accuracy improvement with 69.68% faster training on AG News (Llama3.1-8B), and 76.70% accuracy on GSM8K mathematical reasoning (Qwen2.5-7B) with 42% speedup.

PUBLICATIONS

Peer-Reviewed Conference Papers

- [C1] **Guangyu Meng**, Pengfei Gu, Xueyang Li, Yiyu Shi, Erin W. Chambers, Danny Z. Chen, “TopoAgent: An Agentic Framework for Automated Topology Learning in Medical Imaging,” *European Conference on Computer Vision (ECCV)*, 2026.
- [C2] **Guangyu Meng**, Pengfei Gu, Peixian Liang, John P. Lalor, Erin W. Chambers, Danny Z. Chen, “TopoCL: Topological Contrastive Learning for Medical Imaging,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [C3] **Guangyu Meng**, Qisheng Jiang, Kaiqun Fu, Beiyu Lin, Chang-Tien Lu, Zhiqian Chen, “Early Forecasting of the Impact of Traffic Accidents Using a Single Shot Observation,” *SIAM International Conference on Data Mining (SDM)*, 2022.

Peer-Reviewed Journal Articles

- [J1] **Guangyu Meng**, Qingkai Zeng, John P. Lalor, Hong Yu, “A Psychology-based Unified Dynamic Framework for Curriculum Learning,” *Computational Linguistics*, 2025. DOI: [10.1162/COL.a.584](https://doi.org/10.1162/COL.a.584)
- [J2] Peixian Liang, Yifan Ding, Yizhe Zhang, Jianxu Chen, Hao Zheng, Hongxiao Wang, Yejia Zhang, **Guangyu Meng**, Tim Weninger, Michael Niemier, X. Sharon Hu, Danny Z. Chen, “Cell Instance Segmentation: The Devil Is in the Boundaries,” *IEEE Transactions on Medical Imaging (TMI)*, 2025. DOI: [10.1109/TMI.2025.3621093](https://doi.org/10.1109/TMI.2025.3621093)
- [J3] **Guangyu Meng**, Ruyu Zhou, Liu Liu, Peixian Liang, Fang Liu, Danny Z. Chen, Michael Niemier, X. Sharon Hu, “Efficient Approximation of Earth Mover’s Distance Based on Nearest Neighbor Search,” *IEEE Transactions on Multimedia (TMM)*, 2025. DOI: [10.1109/TMM.2025.3623529](https://doi.org/10.1109/TMM.2025.3623529)

Manuscripts Under Review

- [R1] **Guangyu Meng**, Pengfei Gu, John P. Lalor, Peixian Liang, Erin W. Chambers, Danny Z. Chen, “VTRL-Med: Vision-Topology Reinforced Learning for Medical Image Clustering,” under review at *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027.
- [R2] **Guangyu Meng**, Erin W. Chambers, “Provably Stable Reeb Graph Comparison via Gromov–Wasserstein Distance,” under review at *Journal of Applied and Computational Topology (JACT)*, 2026.
- [R3] **Guangyu Meng**, Pengfei Gu, Erin W. Chambers, Danny Z. Chen, “CuPH: Parallel Cubical Persistent Homology via Level-Synchronous Component Tracking,” under review at *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [R4] **Guangyu Meng**, Mingzhe Li, Erin W. Chambers, “MS-COOT: Comparing Morse–Smale Complexes with Co-Optimal Transport,” under review at *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2026.

PRESENTATIONS

- “A Psychology-based Unified Dynamic Framework for Curriculum Learning,” Workshop on Natural Language Generation, Evaluation, and Metrics (GEM) at ACL 2026, San Diego, CA, July 2026. (Poster)
- “Reeb Graph Comparison via the Gromov–Wasserstein Distance,” Computational Geometry: Young Researchers Forum (CG:YRF 2026), Rutgers University, New Brunswick, NJ, June 2026. (Oral talk)
- “TopoAgent: An Agentic Framework for Automated Topology Learning in Medical Imaging,” Workshop on Foundations of Computational Geometry and Topology, ICERM, Brown University, May 2026. (Poster)
- “TopoAgent: An Agentic Framework for Automated Topology Learning in Medical Imaging,” Mini-Symposium on AI Foundations, University of Notre Dame, May 2026. (Oral talk)
- “Provably Stable Reeb Graph Comparison via Gromov–Wasserstein Distance,” Conference on Topological Data Analysis (TDA 2025), University of Missouri-Columbia, November 2025. (Oral talk and poster)

TEACHING EXPERIENCE

Teaching Assistant

University of Notre Dame

Department of Computer Science and Engineering

2021 – Present

- CSE 60111: Complexity and Algorithms, Spring 2025
- CSE 20110: Discrete Mathematics, Fall 2024 – Designed two course assignments
- CSE 40625/60625: Machine Learning, Fall 2021 – Delivered a tutorial on PyTorch
- CSE 30321: Computer Architecture, Spring 2021

MENTORING

Xianshi Ma (xma3@nd.edu), Ph.D. student, University of Notre Dame: Model fusion for semi-supervised medical video segmentation. *Co-advised with Prof. Danny Z. Chen.*

Colby Nelson (cnelso23@nd.edu), Master’s student, University of Notre Dame: Deep clustering for medical imaging. *Guided project development from problem formulation through experimental design and implementation.*

TECHNICAL SKILLS

Core Domains: Artificial Intelligence, Medical Image Analysis, Semiconductor Verification (EDA)

Deep Learning: PyTorch, TensorFlow, LLM Fine-tuning (LoRA/QLoRA), Transformers, SAM, Vision Transformers

LLM & Agentic AI: Hugging Face, LangChain, LangGraph, Prompt Engineering, Tool-Augmented LLM Agents

Mathematics: Topological Data Analysis (TDA), Persistent Homology, Optimal Transport

Programming: Python, C/C++, CUDA (High-Performance GPU Computing), MATLAB, SystemVerilog

Systems & Tools: Linux/HPC (Slurm), Docker, Git, Mixed-Precision Training, Multi-GPU Distributed Training

PROFESSIONAL SERVICE

Conference Reviewer: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, ACM International Conference on Multimedia (ACM MM) 2026, Workshop on Natural Language Generation, Evaluation, and Metrics (GEM) at ACL 2026, Symposium on Computational Geometry (SoCG) 2025, AAAI Conference on Artificial Intelligence 2024, SIAM International Conference on Data Mining (SDM) 2023

Journal Reviewer: Computing in Geometry and Topology (CGT)

INDUSTRY EXPERIENCE

Emulation Specialist

Mentor Graphics Co., Ltd (now Siemens EDA)

Shenzhen, China

2016 – 2019

- Led high-performance hardware emulation for Huawei’s first commercial 5G chipset; reduced testing runtime from 45 to 8 minutes through parallelization and optimized test code.

ASIC Design Engineer

Spreadtrum Communications

Shanghai, China

2015 – 2016

- Developed Python-based resource allocation tools, improving device utilization by 20%.

NPI Engineer

NXP Semiconductors

Shanghai, China

2013 – 2015

- Optimized wafer mapping and packaging processes, enhancing die bond machine capacity by 14.8%.

REFERENCES

Erin W. Chambers (Co-advisor)

Snyder Family Mission Collegiate Professor
Department of Computer Science and Engineering
University of Notre Dame
echambe2@nd.edu

X. Sharon Hu (Collaborator)

Leo E. and Patti Ruth Linbeck Professor
Department of Computer Science and Engineering
University of Notre Dame
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John P. Lalor (Collaborator)

Assistant Professor, Mendoza College of Business
Concurrent Professor, Computer Science and Engineering
University of Notre Dame
john.lalor@nd.edu

Danny Z. Chen (Co-advisor)

Professor, IEEE Fellow, ACM Distinguished Scientist
Department of Computer Science and Engineering
University of Notre Dame
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Pengfei Gu (Collaborator)

Assistant Professor
Department of Computer Science
University of Texas Rio Grande Valley
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